

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1: FLUIDS

TRINITY TERM 2021

Friday, 11 June

Opening time: 09.30 am UK Time

**You have two hours writing time to complete the paper
and up to 30 minutes technical time to upload your answer file.**

*Answer **two** questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspira during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Page 2 contains physics formulae and data for this paper.

The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible, fluid flow under gravity is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} + \frac{1}{\rho} \nabla p + g \mathbf{k} = \nu \nabla^2 \mathbf{v} ,$$

where $\mathbf{v}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\mathbf{k}$ the vertical unit vector and ν the kinematic viscosity.

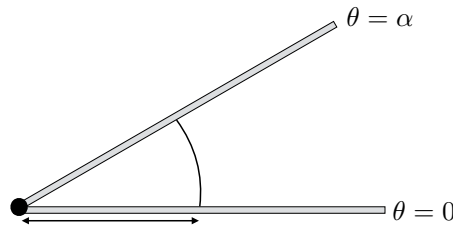
The *Jacobian* $\mathbf{J}$ of a dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is defined as $J_{ij} = \frac{\partial f_i}{\partial x_j}$, where f_i and x_j are the i^{th} and j^{th} components of $\mathbf{f}(\mathbf{x})$ and $\mathbf{x}$ respectively. The trace τ of the Jacobian is equal to the sum of its eigenvalues, while the determinant Δ is equal to their product.

$$1000 \text{ mb} = 1000 \text{ hPa} = 10^5 \text{ Pa}.$$

1. (a) Explain how the physics differs for high Reynolds number fluid flow versus low Reynolds number flow. If the kinematic viscosity of water is $\nu = 10^{-6} \text{m}^2 \text{s}^{-1}$, estimate whether a human swims through water at high or low Reynolds number. [3]

(b) Consider two-dimensional flow characterised by a streamfunction ψ expressed in terms of polar co-ordinates (r, θ) , with velocity components $v_r = (\partial\psi/\partial\theta)/r$ and $v_\theta = -\partial\psi/\partial r$. Show that the flow is incompressible. By considering the curl of the momentum equation, or otherwise, show that the streamfunction for low Reynolds number flow with constant density ρ and constant viscosity μ satisfies $\nabla^4\psi \approx 0$. [4]

(c) Now consider the flow between two long and wide hinged metal plates at $\theta = 0$ and $\theta = \alpha$, immersed in a large bath of viscous fluid, as shown in the figure. The lower plate is fixed at $\theta = 0$. The hinge slowly opens at a rate with $d\alpha/dt$ constant, and the vertex is pinned at $r = 0$. The flow is steady and gravity is negligible. Neglecting end effects, write down the boundary conditions on the flow between the two plates.



The biharmonic equation $\nabla^4\psi = 0$ admits solutions of the form

$$\psi = \sum_{n=1}^{\infty} r^n [A_n \cos n\theta + B_n \sin n\theta + C_n \theta \cos n\theta + D_n \theta \sin n\theta]$$

that are bounded at the origin. Show that

$$\psi = -\frac{\Omega r^2}{2} \left[\frac{(4\alpha \sin 2\alpha)(\theta \sin 2\theta) - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\theta - 2\theta \cos 2\theta)}{4\alpha^2 \sin^2 2\alpha - (\sin 2\alpha + 2\alpha \cos 2\alpha)(\sin 2\alpha - 2\alpha \cos 2\alpha)} \right]$$

is a solution characterising flow between the plates for some Ω to be determined. Determine 2 possible characteristic lengthscales l_1 and l_2 where this solution breaks down (i.e. the solution is valid for r much smaller than the minimum of l_1 and l_2). [9]

(d) Calculate the volume flux crossing a surface with radius r in polar co-ordinates, between the two plates. Physically interpret the resulting value. [3]

(e) Now consider the case where the vertex is no longer pinned and is free to move, and the plates have finite radial length L . Without detailed calculation, explain how the pressure field $p(r, \theta)$ could be determined between the hinged plates. Using a scale estimate or otherwise, explain why the normal force per unit width on the plates $F \propto \mu L d\alpha/dt$.

Discuss what the above solution tells us about the possibility of generating propulsion and motion of the hinge if a motor cyclically opens and closes the hinge over time with $\alpha(t)$ a periodic function of time. [6]

2. (a) With the aid of suitable sketches, explain the physical mechanism that gives rise to propagating sound waves after a small region of a compressible fluid is compressed. [4]

(b) Compressible air flows through a wind tunnel with constant background velocity $\mathbf{v} = U_0 \hat{\mathbf{x}}$, uniform background pressure p_0 and density ρ_0 . The air has equation of state $p = p_0(\rho/\rho_0)^\gamma$ where the pressure is p , density is ρ , and γ is constant. Consider small one dimensional deviations to the background state of the form

$$p = p_0 + p_1(x, t), \quad \rho = \rho_0 + \rho_1(x, t), \quad \mathbf{v} = [U_0 + v_1(x, t)] \hat{\mathbf{x}}.$$

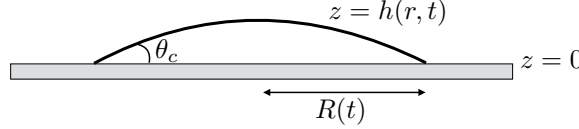
Assume that the air can be treated as effectively inviscid, and neglect gravity. By linearising the equations for conservation of mass and momentum, and the equation of state about the background state, show that the density perturbations satisfy

$$\left[\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x} \right] \left[\frac{\partial}{\partial t} + U_0 \frac{\partial}{\partial x} \right] \rho_1 = c_0^2 \frac{\partial^2 \rho_1}{\partial x^2}, \quad \text{with} \quad c_0^2 = \left. \frac{dp}{d\rho} \right|_{\rho=\rho_0}. \quad [7]$$

(c) A loudspeaker at the centre of the wind tunnel generates density anomalies of the form $\rho_1 = \text{Re}[\rho_0 A \exp(-i\omega t)]$ at $x = 0$ for constant amplitude A and angular frequency ω . Assume that the wave propagation is effectively one-dimensional as considered in part (b), and the ends of the tunnel are open with no wave reflection. Determine a dispersion relation linking ω to c_0 , U_0 and the wavenumber k . Sketch $p_1(x, t)$, $\rho_1(x, t)$ and $v_1(x, t)$ for the special case with $U_0 = 0$, at a time with $\omega t = 2n\pi$ after the waves have propagated the length of the tunnel, where n is an integer. [8]

(d) Two observers at $x = L$ and $x = -L$ are able to continuously measure density $\rho(x, t)$ over an interval of length l , with $2\pi/k < l \ll L$. Describe and explain how their observations differ if $U_0 < c_0$. What do each of the observers see if $U_0 > c_0$? Discuss the physical realism of this solution as $U_0 \rightarrow c_0$. [6]

3. (a) Consider an axisymmetric drop of incompressible fluid resting on a surface, with maximum radius $R(t)$ and thickness $h(r, t)$, as illustrated in the figure. The fluid has constant density ρ , constant viscosity μ , constant surface tension γ , and the gravitational acceleration is g . The atmosphere above the drop can be approximated as inviscid, and with constant pressure p_0 above the drop surface.



Explain why for sufficiently small h the Navier-Stokes equations can be approximated by

$$0 \approx \mu \frac{\partial^2 \mathbf{v}}{\partial z^2} - \nabla p - \rho g \hat{\mathbf{z}}.$$

Using a scaling argument or otherwise, justify that vertical velocities are small in magnitude compared to horizontal velocities for a thin film.

Assume that surface tension results in a pressure jump across the interface between air and liquid given by $p|_-^+ \approx -\gamma\kappa$, where $\kappa(r, t)$ is the curvature of the surface. Determine the pressure $p(r, z, t)$ inside the drop in terms of r, z, γ, κ, h and g .

Hence show that the horizontal velocity in the drop satisfies

$$\mathbf{v} \approx \left[\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right] \frac{z(z-2h)}{2\mu} \hat{\mathbf{r}}. \quad [10]$$

(b) Calculate the volume flux of fluid across a cylindrical surface of radius r . By considering the volume of fluid in the annulus with radius in the interval $[r, r + dr]$ for infinitesimal dr , justify why the film thickness satisfies

$$\frac{\partial h}{\partial t} = \frac{1}{r} \frac{\partial}{\partial r} \left[r \frac{h^3}{3\mu} \left(\rho g \frac{\partial h}{\partial r} + \gamma \frac{\partial \kappa}{\partial r} \right) \right]. \quad [5]$$

(c) Assume that the curvature can be approximated by $\kappa \approx -\partial^2 h / \partial r^2$. The outer edge of the drop contacts the underlying solid surface with constant contact angle θ_c , such that $\partial h / \partial r = -\tan \theta_c$ at $r = R$. There is no cusp at the origin so that $\partial h / \partial r \rightarrow 0$ and all derivatives of h remain finite as $r \rightarrow 0$.

Show that the steady state thickness $h(r)$ of an axisymmetric drop of maximum radius R and volume V , satisfies

$$h = \alpha l \left[\cosh \left(\frac{R}{l} \right) - \cosh \left(\frac{r}{l} \right) \right],$$

where l and α should be determined, and R satisfies

$$1 = \frac{2\pi l^3 \alpha}{V} \int_0^{R/l} \left[\cosh \left(\frac{R}{l} \right) - \cosh(\xi) \right] \xi \, d\xi.$$

Explain the dominant force balances that control the size and shape of the steady state drop. [10]

4. (a) A two-dimensional simple-shear flow has velocity components $v_x = \beta y$, $v_y = 0$ for constant β . Explain why this flow cannot be described using a complex velocity potential. [2]

(b) Determine the streamfunction, and sketch streamlines for flows given by the following complex velocity potentials $w(z)$ with $z = x + iy$:

- (i) $w = \beta z^2$,
- (ii) $w = \beta z^2 e^{i\alpha}$,

where $\beta > 0$ and $0 < \alpha < \pi/2$ are constant. For the flow $w = \beta z^2 e^{i\alpha}$ in part (ii), the circle of fluid $|z| < R$ is marked with dye at time $t = 0$, for constant radius R . Assuming that the dye is purely advected by the flow without significant diffusion, explain how the total area of dyed fluid evolves over time. [8]

(c) A solid impermeable circular object of radius a is introduced into the two-dimensional potential flow from (ii), with centre at $z = 0$ in the complex plane. If the far-field flow is unaffected by the presence of the object, show that the resulting flow can be described by

$$w = Az^2 + \frac{B}{z^2},$$

where A and B should be determined. Hence show that the tangential velocity on the surface of the cylinder has magnitude $v = 4a\beta |\sin(2\theta + \alpha)|$. Explain why the fluid exerts a force on a small element of the surface of the cylinder, and calculate the net force on the object. [8]

(d) Given a real constant $c < a/2$, the Joukowski transform

$$X + iY = Z = z + \frac{c^2}{z},$$

maps a circle with centre at $z = 0$ and radius a , into an ellipse with centre at $Z = 0$, major axis $|X| \leq a + c^2/a$ and minor axis $|Y| \leq a - c^2/a$. Determine the corresponding complex velocity potential $W(Z)$ for flow around the above ellipse, if the flow far from the sphere asymptotes to $W \sim \beta Z^2 e^{i\alpha}$ for $|Z| \gg a$.

N.B. There is no need to algebraically simplify the formula for $W(Z)$ after expressing in closed form.

By considering the transformation of the velocity, or otherwise, determine the location of stagnation points in the Z plane in terms of a , c , α and β . [7]